Institute of Engineering & Technology

Department of Computer Engineering & Application

AI-Powered Smart E-Waste Management & Recycling Intelligence System

A Project Report submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering(AIML)

by

Tumesh Saini (2415500483)

Satyarth Dahiya (2415500420)

Dhairya Goyal (2415500154)

Group No. : 167

Under the Guidance of:

Mr. Preshit Mangesh Desai Signature :

DECLARATION

We **Tumesh Saini , Satyarth Dahiya , Dhairya Goyal** students of B.Tech in Computer Science and Engineering (Artificial Intelligence and Machine Learning) at GLA University, Mathura, hereby solemnly declare that the project report entitled **— AI-Powered Smart E-Waste Management & Recycling Intelligence System** submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology, is an authentic record of our original work carried out during the eighth semester under the valuable guidance and supervision of **Mr. Preshit Mangesh Desai** , Department of Computer Engineering & Applications.

We further declare that the work presented in this report is result of our own efforts, research, and contributions, except where explicit references have been made. The findings, designs, implementations, and conclusions drawn are based on the project conducted by us. This work has not been submitted elsewhere, either in part or in full, for the award of any degree, diploma, or academic qualification.

We also acknowledge that we have adhered to the ethical academic integrity, and project norms set by the University throughout the course of this project.

CERTIFICATE

This is to certify that the project report entitled — **AI-Powered Smart E-Waste Management & Recycling Intelligence System** submitted by Tumesh Saini (2415500483), Satyarth Dahiya (2415500420), Dhairya Goyal (2415500154) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence and Machine Learning) from GLA University, Mathura, is a bona fide record of the original work carried out by them under my supervision and guidance during the academic year 2025–2026.

The work embodied in this report is genuine to the best of my knowledge and has not been submitted, either in part or in full, for the award of any other degree or diploma in this or any other institution. The students have fulfilled all the requirements as per the academic guidelines prescribed by the University.

Mr. Preshit Mangesh Desai

Technical Trainer
Department of Computer
Engineering & Applications GLA
University, Mathura

ACKNOWLEDGEMENT

The successful completion of any project is never the work of a single individual. It is the culmination of collective effort, guidance, and support from many quarters. We take this opportunity to express our profound gratitude to all those who have contributed directly or indirectly to the successful execution of this project.

First and foremost, we extend our deepest gratitude to our project guide, Mr. Preshit Mangesh Desai , Technical Trainer , Department of Computer Engineering & Applications, for her continuous guidance, constructive criticism, and immense patience throughout the duration of this work. His deep knowledge of artificial intelligence, generous sharing of ideas, and constant encouragement have been the cornerstone of this project.

We are also grateful to the Head of Department and all faculty members for providing the necessary infrastructure and academic environment. Finally, we thank our parents and friends for their unwavering moral support.

ABSTRACT

Electronic waste (E-Waste) has emerged as one of the fastest-growing waste streams in the modern world due to rapid technological advancement and increased consumption of electronic devices such as mobile phones, laptops, batteries, and other digital equipment. Improper disposal of e-waste leads to serious environmental pollution, soil contamination, and health hazards due to the presence of toxic substances like lead, mercury, and cadmium.

This project presents an AI-Powered Smart E-Waste Management & Recycling Intelligence System that aims to automate the detection, classification, and sorting of electronic waste using a combination of hardware components and artificial intelligence techniques. The system integrates an IR sensor for object detection, a camera module for capturing waste images, and a machine learning-based classification model to identify different categories of e-waste such as batteries, cables, circuit boards, and mobile devices.

The captured image is processed using computer vision techniques and a trained Convolutional Neural Network (CNN) model, which classifies the waste into predefined categories with high accuracy. Based on the classification result, a microcontroller (Arduino) activates a servo motor to direct the waste into the appropriate recycling bin. This automation significantly reduces human effort and increases sorting efficiency.

Additionally, the system can be extended with IoT capabilities to enable real-time monitoring, data collection, and analysis of waste management patterns. The proposed solution contributes to sustainable waste management practices by improving recycling efficiency, reducing environmental impact, and promoting smart city development.

Overall, this project demonstrates how the integration of Artificial Intelligence, embedded systems, and automation can provide an effective and scalable solution for modern e-waste management challenges.

Keywords :

- E-Waste Management
- Artificial Intelligence (AI)
- Computer Vision
- Convolutional Neural Network (CNN)

Contents :

Chapter 1

1 Introduction :

Electronic waste (E-Waste) is one of the fastest-growing waste streams in the modern world. With the rapid advancement of technology and increased usage of electronic devices such as mobile phones, laptops, batteries, and household electronics, the amount of discarded electronic products is increasing significantly. Improper disposal of these electronic items leads to serious environmental and health issues due to the presence of toxic materials like lead, mercury, and cadmium.

Traditional methods of e-waste management involve manual sorting and disposal, which are inefficient, time-consuming, and harmful to human workers. There is a strong need for an intelligent and automated system that can identify, classify, and sort electronic waste efficiently.

This project proposes an AI-Powered Smart E-Waste Management & Recycling Intelligence System that integrates Artificial Intelligence, Computer Vision, and Embedded Systems to automate the process of e-waste detection and sorting. The system uses sensors to detect the presence of waste, a camera to capture images, and a machine learning model to classify the type of e-waste. Based on the classification result, a microcontroller (Arduino) controls a servo motor to sort the waste into appropriate recycling bins.

The use of AI and automation not only improves efficiency but also reduces human effort and ensures proper recycling of electronic waste. This project contributes to sustainable development and promotes smart waste management practices.

1.1Project Idea :

The main idea of this project is to develop an intelligent system that can automatically detect and classify electronic waste using Artificial Intelligence and hardware components. The system aims to reduce manual effort and improve the efficiency of waste management by using sensors, cameras, and machine learning algorithms.

The smart system captures the image of the waste item, processes it using a trained AI model, and identifies the category of e-waste. Based on this classification, the system automatically directs the waste into the correct recycling bin using a servo motor controlled by an Arduino.

1.2 Motivation of the Project :

The increasing generation of electronic waste and its improper disposal have become major global concerns. Harmful substances present in e-waste can pollute the environment and cause serious health problems. Manual sorting methods are not efficient and expose workers to hazardous conditions.

This project is motivated by the need to create a smart and automated solution that can handle e-waste efficiently. By using Artificial Intelligence and automation, the system reduces human involvement, increases accuracy, and promotes environmentally friendly recycling practices.

The project also aims to contribute towards smart city development and sustainable waste management systems.

1.3 Overview of E-Waste Problem :

Electronic waste (E-Waste) refers to discarded electronic devices such as mobile phones, laptops, televisions, batteries, and other electrical equipment. With the rapid growth of technology and frequent upgrading of devices, the amount of e-waste generated worldwide has increased significantly.

One of the major concerns associated with e-waste is the presence of hazardous materials such as lead, mercury, cadmium, and arsenic. When e-waste is improperly disposed of, these toxic substances can contaminate soil, water, and air, leading to severe environmental pollution and health risks for humans and animals.

Another major issue is the lack of proper recycling systems. In many places, e-waste is either dumped in landfills or processed using unsafe methods, which can release harmful gases and chemicals into the environment. Additionally, manual sorting of e-waste is inefficient and exposes workers to dangerous conditions.

The increasing volume of e-waste and its improper management highlight the urgent need for intelligent and automated solutions. A smart system that can detect, classify, and sort e-waste efficiently can play a significant role in reducing environmental impact and improving recycling processes.

CHAPTER 2

TECHNICAL BACKGROUND :

2.1 Area of Project :

This project falls under the domain of Artificial Intelligence and Embedded Systems. It combines the concepts of Machine Learning, Computer Vision, and hardware-based automation to create a smart e-waste management system.

Artificial Intelligence plays a key role in identifying and classifying electronic waste using image processing techniques. Computer Vision is used to analyze the captured images of waste materials, while Embedded Systems such as Arduino are used to control the hardware components like sensors and motors.

The project also contributes to the field of Smart Waste Management and Smart City Development by providing an efficient and automated solution for handling electronic waste.

2.2 Technical Keywords :

E-Waste Management

Artificial Intelligence (AI)

Computer Vision

Convolutional Neural Network (CNN)

Arduino

2.3 Technologies Used :

The project uses a combination of hardware and software technologies to achieve efficient e-waste management.

Hardware Technologies:

- Arduino Microcontroller: Used for controlling sensors and actuators.
- Infrared Proximity Sensor: Detects the presence of waste by sensing nearby objects without physical contact.
- Servo Motor: Used for sorting waste into different bins.
- Jumper Wire Set: Used for establishing electrical connections between different components in the circuit.

Software Technologies:

- Python: Used for implementing the AI model.

- OpenCV: Used for image processing and computer vision tasks.

- TensorFlow/Keras: Used to build and train the machine learning model.

- Arduino IDE: Used for programming the microcontroller.

These technologies work together to create an intelligent system capable of detecting, classifying, and sorting e-waste automatically.

CHAPTER 3:
PROBLEM DEFINITION AND SCOPE

3.1 Problem Statement :

The rapid increase in electronic device usage has led to a significant rise in electronic waste (e-waste). Improper disposal of e-waste poses serious environmental and health risks due to the presence of toxic substances such as lead, mercury, and cadmium.

Traditional waste management systems rely on manual sorting, which is inefficient, time-consuming, and hazardous for workers. There is a lack of automated systems that can accurately detect and classify e-waste for proper recycling.

Therefore, there is a need for an intelligent system that can automatically identify and sort electronic waste to improve efficiency and reduce environmental impact.

3.2 Goals and Objectives :

The main goal of this project is to develop an AI-based smart system for efficient e-waste management.

Objectives of the project include:

- To detect the presence of e-waste using sensors
- To classify different types of electronic waste using Artificial Intelligence
- To automate the sorting process using a microcontroller and servo motor
- To reduce manual effort and improve accuracy
- To promote environmentally friendly recycling practices

3.3 Scope of the Project :

The scope of this project is limited to the detection, classification, and sorting of electronic waste using a prototype system.

The system is designed to handle common e-waste items such as batteries, cables, circuit boards, and small electronic devices. It can be implemented in small-scale environments such as labs, recycling units, or smart bins.

In the future, the system can be expanded to large-scale industrial applications and integrated with IoT and cloud-based technologies for real-time monitoring and data analysis.

3.4 Constraints :

The project has certain limitations and constraints:

- Limited hardware resources and processing power
- Accuracy of the AI model depends on the quality of training data
- Prototype system may not handle all types of e-waste
- Environmental factors like lighting and positioning may affect performance
- Budget constraints for advanced components

CHAPTER 4

METHODOLOGY

4.1 System Working :

The proposed system works by integrating sensors, a microcontroller, and Artificial Intelligence techniques to detect and manage e-waste efficiently.

When an object is placed inside the smart bin, the Infrared Proximity Sensor detects the presence of the object. Once detected, the system initiates the classification process. The classification can be performed using predefined logic or an AI-based model depending on the implementation.

After identifying the type of waste, the Arduino microcontroller sends signals to the servo motor. The servo motor rotates to direct the waste into the appropriate bin for proper recycling.

This process ensures automatic detection and sorting of e-waste with minimal human intervention.

4.2 Data Collection :

Data collection is an essential step for developing the AI-based classification system. The dataset consists of images of various types of electronic waste such as batteries, cables, circuit boards, and mobile devices.

These images are collected from online sources and image datasets. The data is organized into different categories based on the type of e-waste.

Proper labeling is performed to ensure that the AI model can accurately learn and classify different types of waste.

4.3 Image Processing & AI Model :

Image processing techniques are used to prepare the input data for classification. The collected images are resized to a uniform size to maintain consistency.

Techniques such as image normalization, filtering, and enhancement are applied to improve the quality of the images. This helps in extracting important features from the images and improves the performance of the AI model.

These processed images are then used as input for the classification system.

The system uses a Convolutional Neural Network (CNN) as the AI model for image classification. CNN is highly effective in identifying patterns and features in images.

The model is trained on the prepared dataset to recognize different categories of e-waste. During training, the model learns to distinguish between various types of waste based on visual features.

Once trained, the model can classify new input images and provide accurate predictions for waste sorting.

CHAPTER 5

SYSTEM DESIGN

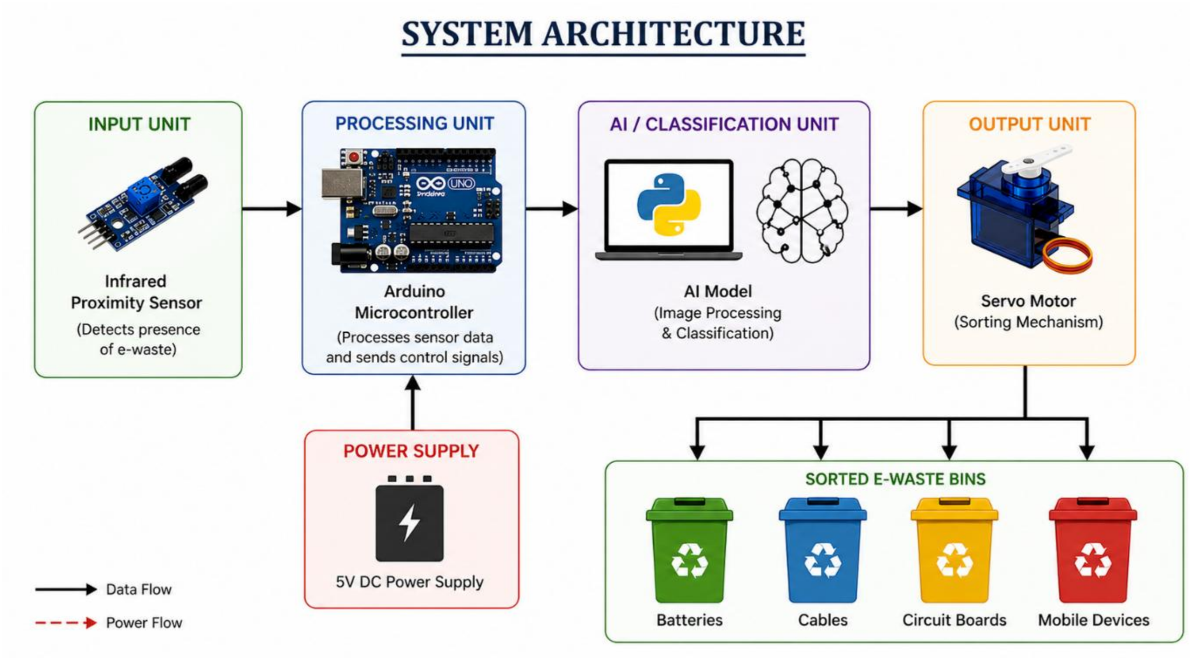
5.1 System Architecture :

The system architecture represents the overall structure and workflow of the proposed smart e-waste management system. It shows how different components such as sensors, microcontroller, and actuators interact with each other.

In this system, the Infrared Proximity Sensor detects the presence of e-waste. Once the object is detected, the system processes the input and determines the type of waste based on predefined logic or AI-based classification.

The Arduino microcontroller acts as the central processing unit, which receives input signals from the sensor and controls the servo motor accordingly. The servo motor is responsible for directing the waste into the appropriate bin.

This architecture ensures efficient communication between hardware and software components, resulting in an automated and reliable waste management system.



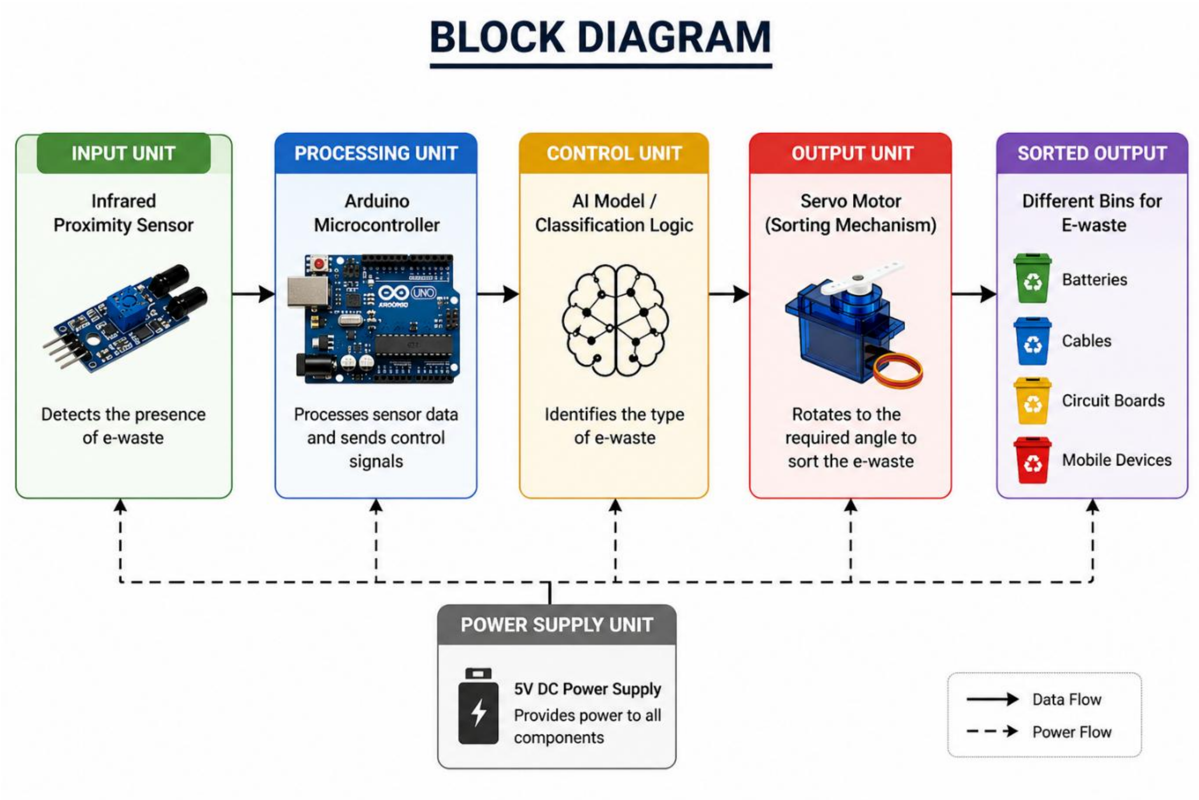
5.2 Block Diagram :

The block diagram of the system illustrates the functional components and their interconnections.

The major components include:

- Input Unit: Infrared Proximity Sensor used for detecting the presence of waste.
- Processing Unit: Arduino microcontroller which processes the input signals.
- Control Unit: Logic or AI model that determines the type of waste.
- Output Unit: Servo motor that performs the sorting action.

The data flow starts from the sensor, passes through the processing unit, and ends at the output unit where the waste is sorted into the correct bin.



5.3 Flowchart :

The flowchart represents the step-by-step working of the system.

Step 1: Start

Step 2: Detect object using Infrared Proximity Sensor

Step 3: Check if object is present

Step 4: If No → Wait for input

Step 5: If Yes → Identify type of waste

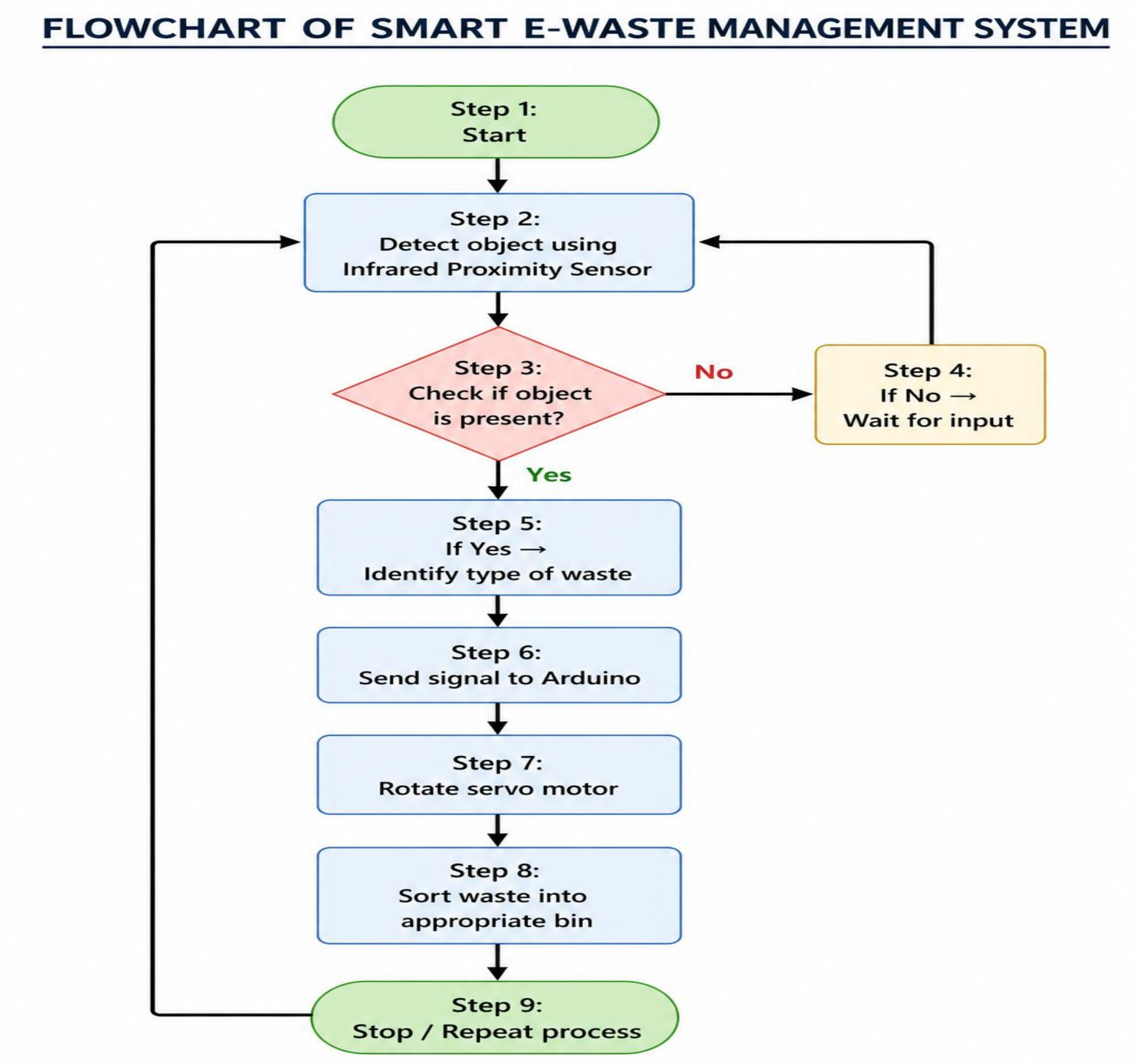
Step 6: Send signal to Arduino

Step 7: Rotate servo motor

Step 8: Sort waste into appropriate bin

Step 9: Stop / Repeat process

The flowchart clearly shows the logical sequence of operations in the system.



CHAPTER 6

IMPLEMENTATION

6.1 Introduction :

The implementation phase focuses on the practical development of the AI-Powered Smart E-Waste Management System. It involves the integration of hardware components such as sensors, Arduino, and servo motor along with the software logic required for automation.

The system is designed to detect the presence of e-waste, process the input, and perform sorting automatically. Both hardware and software components work together to achieve efficient waste management.

6.2 Hardware Implementation :

The hardware implementation consists of assembling and connecting all physical components required for the system.

The Arduino microcontroller is used as the central unit that controls all operations. The Infrared Proximity Sensor is used to detect the presence of an object inside the bin. Once the object is detected, the Arduino processes the signal and activates the servo motor.

The servo motor is responsible for sorting the waste into different bins by rotating at specific angles. All components are connected using jumper wires and mounted on a breadboard for proper circuit design.

A stable power supply is provided to ensure smooth functioning of the system. The hardware setup is simple, cost-effective, and easy to implement.

6.3 Software Implementation :

The software implementation involves writing the program that controls the working of the system.

The Arduino is programmed using Arduino IDE. The code reads input from the Infrared Proximity Sensor and processes it to determine whether an object is present. Based on predefined conditions or classification logic, the Arduino sends signals to the servo motor.

If AI-based classification is used, the model is developed using Python and trained using machine learning techniques. The output of the model is used to decide the movement of the servo motor.

The software ensures proper coordination between input detection and output action.

6.5 Algorithm :

Step 1: Start

Step 2: Detect object using Infrared Proximity Sensor

Step 3: Check if object is present

Step 4: If No → Wait for input

Step 5: If Yes → Identify type of waste

Step 6: Send signal to Arduino

Step 7: Rotate servo motor

Step 8: Sort waste into appropriate bin

Step 9: Stop / Repeat process

CHAPTER 7

RESULTS AND TESTING

7.1 Test Cases :

The system was tested with different types of electronic waste to check its functionality.

Test Case 1:

Input: Battery

Expected Output: Sorted into battery bin

Result: Successful

Test Case 2:

Input: Cable

Expected Output: Sorted into cable bin

Result: Successful

Test Case 3:

Input: Circuit Board

Expected Output: Sorted into circuit board bin

Result: Successful

Test Case 4:

Input: Mobile Device

Expected Output: Sorted into mobile bin

Result: Successful

Test Case 5:

Input: No object present

Expected Output: System remains idle

Result: Successful

7.2 Results :

The system successfully detected and sorted electronic waste into appropriate bins. The Infrared Proximity Sensor accurately detected the presence of objects, and the Arduino effectively controlled the servo motor.

The sorting mechanism worked efficiently, and the system was able to perform continuous operations without failure. The results indicate that the system is reliable and suitable for small-scale implementation.

7.4 Performance Analysis :

The performance of the system was evaluated based on accuracy, efficiency, and response time.

- Accuracy: The system showed high accuracy in detecting and sorting e-waste correctly.
- Efficiency: The automated system reduced manual effort and improved sorting speed.
- Response Time: The system responded quickly after detecting the object.

Overall, the system performed well under testing conditions and achieved the desired objectives.

CHAPTER 8
APPLICATIONS
8.1 Applications of the System :

The proposed system can be applied in the following areas:

- Smart Cities: The system can be integrated into smart waste management systems to automatically sort electronic waste and improve urban cleanliness.

- Recycling Centers: It can be used in recycling plants to reduce manual sorting effort and increase efficiency.

- Educational Institutions: The system can be used as a learning tool in colleges and schools to demonstrate the application of AI and embedded systems.

- Industrial Applications: Small-scale industries can use this system for managing their electronic waste efficiently.

- Public Places: Smart bins equipped with this system can be installed in public areas such as malls, offices, and parks for proper e-waste disposal.

CHAPTER 9

CONCLUSION AND FUTURE SCOPE

9.1 Conclusion :

The AI-Powered Smart E-Waste Management & Recycling Intelligence System successfully demonstrates an efficient and automated solution for handling electronic waste. The system integrates Artificial Intelligence, Computer Vision, and Embedded Systems to detect and sort e-waste with minimal human intervention.

The use of an Infrared Proximity Sensor ensures accurate detection of objects, while the Arduino microcontroller effectively controls the sorting mechanism using a servo motor. The implementation of AI-based classification enhances the intelligence of the system and improves decision-making capabilities.

The system reduces manual effort, increases efficiency, and promotes environmentally friendly waste management practices. It provides a practical solution for small-scale applications and contributes towards sustainable development and smart city initiatives.

Overall, the project achieves its objectives and proves to be a reliable and effective approach for managing electronic waste.

9.2 Future Scope :

The proposed system can be further improved and expanded in the future with advanced technologies and enhancements.

- Integration with IoT for real-time monitoring and data analysis
- Development of a mobile application for remote control and tracking
- Use of advanced AI models for higher classification accuracy
- Implementation in large-scale industrial recycling systems
- Addition of more sensors for detecting different types of waste
- Cloud-based data storage and analytics for better decision-making

These improvements can make the system more efficient, scalable, and suitable for real-world applications.

REFERENCES :

[1] Arduino Official Website, https://www.arduino.cc

[2] Arduino Servo Motor Documentation, https://www.arduino.cc/reference/en/libraries/servo/

[3] Infrared Proximity Sensor Working, Electronics Tutorials

[4] OpenCV Documentation for Image Processing, https://docs.opencv.org

[5] TensorFlow/Keras Documentation for CNN Models, https://www.tensorflow.org

[6] Kaggle Dataset for E-Waste Image Classification, https://www.kaggle.com

[7] United Nations Environment Programme (UNEP), "Global E-Waste Monitor Report"

[8] Research Paper: “E-Waste Management using Machine Learning Techniques”

[9] Research Paper: “Smart Waste Management System using IoT and Sensors”

PROJECT PLANNER: